

Technical Report

English Proficiency Across Europe and Italy Datasets, Preprocessing, and Methodological Considerations

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1 Introduction

This technical report documents the datasets, preprocessing procedures, and methodological considerations underlying the development of the interactive data visualization project initially introduced in the project proposal.

The project explores English proficiency patterns across Europe and Italy through the integration of multiple heterogeneous public datasets related to English proficiency, educational exposure, regional socio-economic indicators, and educational quality metrics. The primary objective of the visualization system is exploratory: to identify spatial disparities, temporal trends, and potential associations between English proficiency outcomes and selected contextual variables.

While the initial proposal defined the general analytical framework and the intended visualization objectives, the implementation process required several methodological adaptations related to data availability, harmonisation constraints, missing observations, and differences in temporal and spatial granularity across datasets. Consequently, certain aspects of the final visualization system differ from the original proposal, particularly regarding variable selection, preprocessing procedures, and visualization scope.

The final project was developed as a web-based interactive dashboard combining choropleth maps, comparative visualizations, temporal trend analysis, and exploratory filtering tools. The dashboard additionally includes interactive year selection, comparative country rankings, European aggregate indicators, and dedicated comparisons between Italy and broader European trends. Particular attention was given to dataset consistency, temporal alignment strategies, variable standardisation, and the treatment of incomplete observations.

Because the project combines datasets originating from different institutions, years, and reporting methodologies, substantial preprocessing and harmonisation operations were required before visualization development. The purpose of this technical report is therefore to document the structure and quality of the datasets used, the preprocessing operations performed, and the methodological limitations that may affect interpretation of the visualized results.

Each section of the report corresponds to one component of the final visualization project and presents the associated datasets, transformations, descriptive insights, and limitations.

2 Part 1 — ESL Exposure and English Proficiency Across Europe

2.1 Dataset Origin, Structure, Variables, and Metadata

This section investigates the relationship between exposure to English as a second language (ESL) in upper-secondary education and later English proficiency outcomes across European countries.

Two primary datasets were integrated for this sub-project.

The first dataset is the *EF English Proficiency Index* (EF EPI), published annually by Education First. The dataset contains country-level English proficiency scores derived from voluntary online English tests completed by adult participants. The analysis retained annual observations from 2016 to 2025 for European countries. Main variables include country name, EF EPI score, ranking, and year.

The second dataset originates from Eurostat: *“Pupils by education level and modern foreign language studied”*. This dataset contains yearly statistics describing the proportion and number of pupils studying foreign languages across European educational systems. For this project, only upper-secondary students learning English as a foreign language were retained. Relevant variables include country identifier, year, educational level, language studied, and percentage of pupils learning English.

Because Eurostat datasets were initially structured in wide format, preprocessing included reshaping operations transforming yearly columns into long-format country-year observations suitable for temporal analysis and dataset merging. Additional filtering operations retained only observations corresponding to English language learning indicators, percentage-based measurements, and total population aggregates in order to ensure cross-country comparability.

Before integration, both datasets required substantial preprocessing and harmonisation. Country names and identifiers were standardised to ensure consistency across sources. Additional harmonisation rules were implemented for specific naming inconsistencies (e.g., “Czech Republic” to “Czechia”, “Türkiye” to “Turkey”, and “UK” to “United Kingdom”). ISO-3 country codes were subsequently generated and used as the primary integration key across datasets and dashboard visualisations. Observations with incomplete or missing country-year combinations were filtered during preprocessing.

To account for delayed educational effects, a temporal lag alignment strategy was implemented. ESL exposure measured at year t was associated with EF EPI proficiency outcomes measured at year $t + 4$. This lag structure was introduced to approximate the delayed impact of secondary-school English exposure on later adult proficiency outcomes.

Additionally, percentile standardisation was applied yearly in order to improve comparability between variables measured on different scales. A derived variable, `gap_pct`, was constructed as:

$$\text{gap_pct} = \text{EF percentile} - \text{ESL exposure percentile}$$

Positive values indicate countries performing above expected educational exposure levels, whereas negative values indicate relative underperformance.

After preprocessing, harmonisation, and temporal lag alignment, the final analytical dataset retained a reduced subset of European countries with sufficiently complete observations across both datasets.

Dataset	Years	Level	Main Variables
EF EPI	2016–2025	Country	EF score, ranking
Eurostat ESL	2012–2021	Country	ESL exposure rate

Table 1: Main datasets integrated for Part 1 of the project.

Figure 1 illustrates the spatial distribution of the derived `gap_pct` variable across European countries for the latest available year after preprocessing and dataset harmonisation. Certain

countries are absent due to missing or incomplete country-year observations removed during preprocessing and temporal lag alignment.

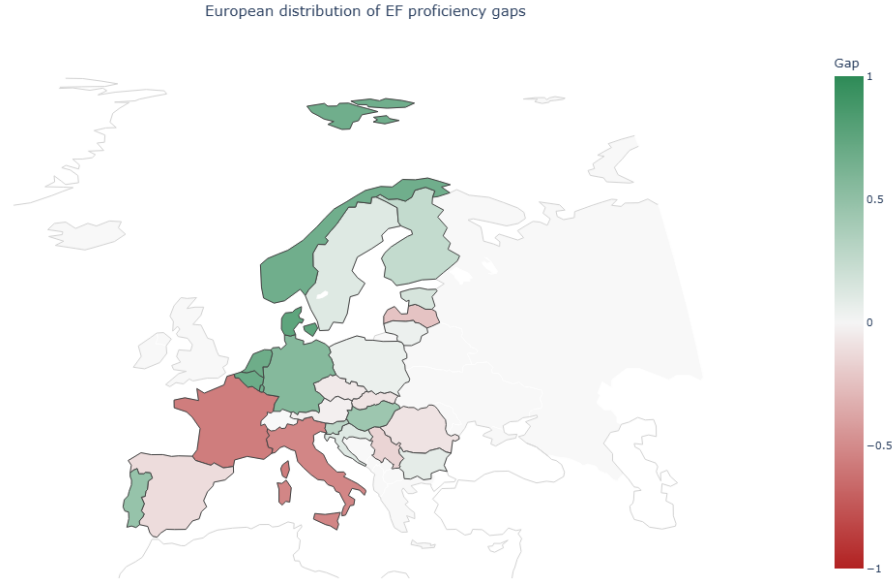


Figure 1: Percentile-based proficiency gap across European countries for the latest available year after preprocessing and temporal lag alignment. Positive values indicate countries whose EF EPI percentile exceeds their ESL exposure percentile, whereas negative values indicate relative underperformance.

2.2 Descriptive Statistics and Initial Insights

Exploratory analysis revealed persistent spatial disparities in English proficiency across Europe. Northern European countries consistently occupied the highest positions in EF EPI rankings, while several Southern and Eastern European countries displayed lower proficiency levels.

Temporal evolution appeared relatively stable throughout the observation period. Although certain countries exhibited moderate improvements or declines, the overall European hierarchy remained comparatively stable between 2016 and 2025.

The percentile-based comparison highlighted notable deviations between educational exposure and observed proficiency outcomes. Certain countries achieved relatively strong proficiency despite lower-than-expected exposure indicators, whereas others underperformed relative to their educational exposure levels.

Interactive choropleth maps and temporal trend visualisations further revealed regional clustering patterns and long-term stability in proficiency disparities across Europe.

2.3 Identification of Missing Data, Noise, Bias, Inconsistencies, and Limitations

Several methodological limitations affect the interpretation of the results.

First, EF EPI data is based on voluntary online participation rather than probabilistic sampling. This introduces self-selection bias, since participants are more likely to be individuals already interested in testing their English proficiency. Consequently, EF EPI scores may not perfectly represent national populations.

Second, comparability across countries may be partially affected by heterogeneous participation rates, internet access, and demographic differences between respondents. Although the

analysis assumes that these biases remain relatively stable across countries and years, this assumption cannot be fully verified.

Third, the observational nature of the project prevents causal inference. The visualisations and comparisons only identify associations and relative patterns between ESL exposure and proficiency outcomes.

Additional limitations arise from temporal disruptions linked to the COVID-19 pandemic, which may have affected both educational exposure and language acquisition dynamics during specific years.

Finally, missing observations and inconsistencies across datasets required preprocessing decisions, including filtering incomplete country-year combinations and harmonising country identifiers across data sources. Although these operations improved dataset consistency, they reduced the number of usable observations for certain years.

3 Conclusion

The development of this project required the integration and harmonisation of multiple heterogeneous datasets originating from different public institutions, reporting systems, and spatial scales. Across all sections of the project, substantial preprocessing operations were necessary, including data cleaning, country and regional harmonisation, temporal alignment, variable standardisation, and the treatment of incomplete observations.

The implementation process also highlighted several practical challenges commonly encountered in data visualization projects involving heterogeneous open datasets. Differences in temporal coverage, missing observations, varying spatial granularity, and inconsistencies in reporting methodologies required methodological adaptations compared with the initial project proposal.

Because the project remains exploratory and observational, important methodological limitations persist. Several datasets contain potential sampling biases, heterogeneous coverage, or measurement inconsistencies across years and regions. Consequently, the visualizations should primarily be interpreted as descriptive and exploratory tools rather than instruments for causal inference.

Despite these limitations, the project demonstrates how interactive visualization techniques can support the exploration of complex educational and socio-economic patterns across Europe and Italy. The final dashboard enables users to compare regions, investigate temporal trends, and explore potential relationships between English proficiency indicators and broader contextual variables through an accessible visual interface.

More broadly, the project highlights both the analytical potential and the practical challenges associated with combining heterogeneous public datasets within a unified interactive data visualization framework.

The complete reference list and broader theoretical context are available in the accompanying project proposal document.